

OSSP_WEEK_1_SKILL

Step 1 — Open Ubuntu Terminal

Press: Ctrl + Alt + T

You should see something similar to:

Step 2 — Update Ubuntu

Run:

```
sudo apt update
```

Enter your Ubuntu password when asked.

Then run:

```
sudo apt upgrade -y
```

Wait until the commands finish.

Step 3 — Install GCC, Make, Git, strace and pstree

Run:

```
sudo apt install build-essential git strace psmisc -y
```

```
Teja@LAPTOP-TJEGMFM8:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.00
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
Teja@LAPTOP-TJEGMFM8:~$ git --version
git version 2.43.0
Teja@LAPTOP-TJEGMFM8:~$ make --version
GNU Make 4.3
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2020 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Teja@LAPTOP-TJEGMFM8:~$ strace -V
strace -- version 6.8
Copyright (c) 1991-2024 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx32-mpers
Teja@LAPTOP-TJEGMFM8:~$ pstree --version
pstree (PSmisc) 23.7
Copyright (C) 1993-2024 Werner Almesberger and Craig Small

PSmisc comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it under
the terms of the GNU General Public License.
For more information about these matters, see the files named COPYING.
Teja@LAPTOP-TJEGMFM8:~$ |
```

Verify GCC:

```
gcc --version
```

Verify Git:

```
git --version
```

Verify Make:

```
make --version
```

Verify strace:

```
strace -V
```

Verify pstree:

```
pstree -version
```

Step 4 — Configure Git

Set your name:

```
git config --global user.name "Your Name"
```

Example:

```
git config --global user.name "Laxman Reddy"
```

Set your email:

```
git config --global user.email "your-email@example.com"
```

Check:

```
git config --global --list
```

```
Teja@LAPTOP-TJEGMFM8:~$ git config --global user.name "Teja"
Teja@LAPTOP-TJEGMFM8:~$ git config --global user.email "srivarunpot@gmail.com"
Teja@LAPTOP-TJEGMFM8:~$ git config --global --list
user.name=Teja
user.email=srivarunpot@gmail.com
Teja@LAPTOP-TJEGMFM8:~$ |
```

Step 5 — Create Your Project Folder

```
mkdir -p ~/ossp_process_lab
```

```
cd ~/ossp_process_lab
```

Check location:

```
Pwd
```

```
Teja@LAPTOP-TJEGMFM8:~$ mkdir -p ~/ossp_process_lab
Teja@LAPTOP-TJEGMFM8:~$ cd ~/ossp_process_lab
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ pwd
/home/Teja/ossp_process_lab
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/Teja/ossp_process_lab/.git/
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ |
```

```
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ mkdir -p src include build
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ touch README.md Makefile src/process_demo.c
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ find . -maxdepth 2 -print
.
./src
./src/process_demo.c
./include
./README.md
./.git
./.git/HEAD
./.git/branches
./.git/config
./.git/description
./.git/hooks
./.git/objects
./.git/info
./.git/refs
./build
./Makefile
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ |
```


Step 6 — Create Git Repository

```
git init
```

Step 7 — Create Project Structure

```
mkdir -p src include build
```

```
touch README.md Makefile src/process_demo.c
```

Check:

```
find . -maxdepth 2 -print
```

Step 8 — Check the Shell

```
echo $SHELL
```

```
ps -p $$ -f
```

```
pstree -p $$
```

```
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ echo $SHELL
/bin/bash
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ ps -p $$ -f
UID          PID    PPID  C STIME TTY          TIME CMD
Teja          480     479   0 04:52 pts/0        00:00:00 -bash
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ pstree -p $$
bash(480)——pstree(12809)
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ nano src/process_demo.c
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ nano Makefile
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ make
gcc -Wall -Wextra src/process_demo.c -o build/process_demo
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ ls -l build/
total 16
-rwxr-xr-x 1 Teja Teja 16304 Aug 24 05:04 process_demo
```


Step 9 — Create the C Program

Open the source file:

```
nano src/process_demo.c
```

Paste this exact program:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }
    else if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        printf("Child Parent PID: %d\n", getppid());

        execlp("ls", "ls", "-l", NULL);

        perror("exec failed");
        exit(1);
    }
    else
    {
        printf("Parent is waiting for child...\n");
        wait(NULL);
        printf("Parent: child completed.\n");
    }

    return 0;
}
```

Save: Ctrl + O → Enter → Ctrl + X

Step 10 — Create the Makefile

```
nano Makefile
```

Paste:

```
CC = gcc
CFLAGS = -Wall -Wextra
TARGET = build/process_demo
SOURCE = src/process_demo.c

all: $(TARGET)

$(TARGET): $(SOURCE)
    $(CC) $(CFLAGS) $(SOURCE) -o $(TARGET)

run: $(TARGET)
    ./$(TARGET)

clean:
    rm -f $(TARGET)
```

IMPORTANT: The command lines in the Makefile must begin with a TAB, not spaces.

Save: Ctrl + O → Enter → Ctrl + X

Step 11 — Compile Using Make

```
make
ls -l build/
```

Step 12 — Run the Program

```
./build/process_demo
```

Step 13 — Run Using the Makefile

```
make run
```

Step 14 — Check the Process Tree

```
pstree -p
ps --forest -f
```

Step 15 — Observe the Parent-Child Relationship

```
./build/process_demo
```

Step 18 — Run strace

```
strace -f ./build/process_demo
```

Step 19 — Trace Only Process-Related Calls

```
strace -f -e trace=process ./build/process_demo
```

Step 20 — Save strace Output

```
strace -f -e trace=process ./build/process_demo 2> strace_output.txt
```

```
cat strace_output.txt
```

Step 21 — Check Git Status

```
git status
```

Step 22 — Add Project Files

```
git add .
```

```
git status
```

Step 23 — Create Git Commit

```
git commit -m "Initial OSSP process abstraction lab"
```

```
git log --oneline
```

Step 24 — Clean the Build

```
make clean
```

```
ls -l build/
```

Step 25 — Build Again

```
make
```

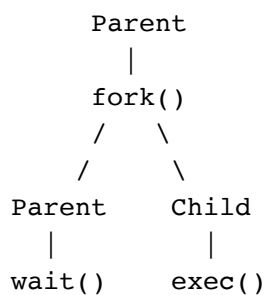
```
ls -l build/
```

Step 26 — Final Run

```
make run
```

Step 16 — Understand fork()

Before `fork()`, there is one process. After `fork()`, there are two processes: the parent and the child.



In the child, `fork()` returns 0. In the parent, `fork()` returns the child's PID. A negative value indicates failure.

Step 17 — Understand `exec()`

The child executes:

```
execlp("ls", "ls", "-l", NULL);
```

The `exec()` call replaces the child process image with the `ls` program. If `exec()` succeeds, statements after it are not executed.

```
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ strace -f -e trace=process ./build/process_demo 2> strace_output.txt
Parent PID: 12877
Parent is waiting for child...
Child PID: 12878
Child Parent PID: 12877
total 20
-rw-r--r-- 1 Teja Teja 220 Aug 24 05:04 Makefile
-rw-r--r-- 1 Teja Teja 6 Aug 24 05:02 README.md
drwxr-xr-x 2 Teja Teja 4096 Aug 24 05:04 build
drwxr-xr-x 2 Teja Teja 4096 Aug 24 05:02 include
drwxr-xr-x 2 Teja Teja 4096 Aug 24 05:03 src
-rw-r--r-- 1 Teja Teja 728 Aug 24 05:07 strace_output.txt
Parent: child completed.
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ cat strace_output.txt
execve("./build/process_demo", ["/build/process_demo"], 0x7fffc4d7ffd68 /* 23 vars */) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARPID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x7f9176844a10) = 12878
[pid 12877] wait4(-1, <unfinished ...>)
[pid 12878] execve("/usr/local/sbin/ls", ["ls", "-l"], 0x7ffd911bc448 /* 23 vars */) = -1 ENOENT (No such file or directory)
[pid 12878] execve("/usr/local/bin/ls", ["ls", "-l"], 0x7ffd911bc448 /* 23 vars */) = -1 ENOENT (No such file or directory)
[pid 12878] execve("/usr/sbin/ls", ["ls", "-l"], 0x7ffd911bc448 /* 23 vars */) = -1 ENOENT (No such file or directory)
[pid 12878] execve("/usr/bin/ls", ["ls", "-l"], 0x7ffd911bc448 /* 23 vars */) = 0
[pid 12878] exit_group(0) = ?
[pid 12878] +++ exited with 0 +++
--- wait4 resumed>NULL, 0, NULL) = 12878
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=12878, si_uid=1000, si_status=0, si_utime=0, si_stime=0} ---
exit_group(0) = ?
+++ exited with 0 +++
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Makefile
    README.md
    build/
    src/
    strace_output.txt

nothing added to commit but untracked files present (use "git add" to track)
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ |
```

```

Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ git add
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   Makefile
    new file:   README.md
    new file:   build/process_demo
    new file:   src/process_demo.c
    new file:   strace_output.txt

Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ git commit -m "Initial OSSP process abstraction lab"
[master (root-commit) 8069063] Initial OSSP process abstraction lab
 5 files changed, 67 insertions(+)
 create mode 100644 Makefile
 create mode 100644 README.md
 create mode 100755 build/process_demo
 create mode 100644 src/process_demo.c
 create mode 100644 strace_output.txt
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ git log --oneline
8099063 (HEAD -> master) Initial OSSP process abstraction lab
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ make clean
rm -f build/process_demo
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ ls -l build/
total 0
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ make
gcc -Wall -Wextra src/process_demo.c -o build/process_demo
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ ls -l build/
total 16
-rwxr-xr-x 1 Teja Teja 16304 Aug 24 05:10 process_demo
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ |

```

```

Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ make run
./build/process_demo
Parent PID: 12931
Parent is waiting for child...
Child PID: 12932
Child Parent PID: 12931
total 20
-rw-r--r-- 1 Teja Teja 220 Aug 24 05:04 Makefile
-rw-r--r-- 1 Teja Teja 6 Aug 24 05:02 README.md
drwxr-xr-x 2 Teja Teja 4096 Aug 24 05:10 build
drwxr-xr-x 2 Teja Teja 4096 Aug 24 05:02 include
drwxr-xr-x 2 Teja Teja 4096 Aug 24 05:03 src
-rw-r--r-- 1 Teja Teja 1039 Aug 24 05:07 strace_output.txt
Parent: child completed.
Teja@LAPTOP-TJEGMFM8:~/ossp_process_lab$ |

```